Fiscal Year:	FY 2018	**Task Last Updated:**	FY 06/01/2018
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Neurostructural, Cognitive, and Physiologic Changes During a 1-year Antarctic Winter-Over Mission		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	University of Pennsylvania		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2014	**End Date:**	03/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2019 per NSSC information (Ed., 1/10/19) NOTE: End date changed to 12/31/2018 per NSSC information (Ed., 8/7/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17) NOTE: Extended to 7/31/2018 per K. Ohnesorge/JSC and NSSC information (Ed., 12/7/16)		
Key Personnel Changes/Previous PI:	May 2017: Nathalie Pattyn (Vrije Universiteit Brussel, Belgium) was added as a co-investigator. She was the research MD on Halley station and will correlate Cognition data with SINSKILL data (also acquired in Halley). May 2016 report: Damien Leger and Gabi Ambrecht were added as Co-Investigators as we are sharing actigraphy data with them. May 2015 report: David Roalf, PhD was added as a Co-Investigator to take over most of the tasks of Ted Sattherthwaite, MD. Alexander Stahn, PhD and Hanns-Christian Gunga, MD PhD were added as Co-Investigators for Cognition in Neumayer-III and Halley-VI. Simone Macri and Mirjam Münch were added as Co-Investigators as we are sharing actigraphy data with them.		

COI Name (Institution):	Bilker, Warren Ph.D. (University of Pennsylvania) Dinges, David Ph.D. (University of Pennsylvania) Elliott, Mark Ph.D. (University of Pennsylvania) Goel, Namni Ph.D. (University of Pennsylvania) Gur, Ruben Ph.D. (University of Pennsylvania) Satterthwaite, Theodore M.D. (University of Pennsylvania) Johannes, Bernd Ph.D. (German Aerospace Center (DLR), Institute of Aerospace Medicine) Mollicone, Daniel Ph.D. (Pulsar Informatics, Inc.) Roalf, David Ph.D. (University of Pennsylvania) Stahn, Alexander Ph.D. (Charité Berlin) Gunga, Hanns-Christian M.D., Ph.D. (Charité Berlin) Macri, Simone Ph.D. (Istituto Superiore di Sanità) Münch, Mirjam Ph.D. (Charité Berlin) Leger, Damien M.D., Ph.D. (Université Paris Descartes) Ambrecht, Gabriele (Charité Berlin) Pattyn, Nathalie M.D., Ph.D. (Vrije Universiteit Brussel, Belgium)
Grant/Contract No.:	NNX14AM81G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This proposal primarily addresses the Behavioral Medicine (BMed) 3 Gap on the nature and duration of cognitive performance changes in-flight and post mission, by assessing neurostructural, cognitive, behavioral, physiologic, and psychosocial changes in maximally N=24-28 crewmembers during a 10-12 month Antarctic winter-over in Concordia station, and in the same number of controls matched to crewmembers based on age, gender, and educational attainment. State-of -the-art quantitative structural and functional magnetic resonance imaging (fMRI), in both resting-state and activated; diffusion tensor imaging; and arterial spin labeled fMRI will be performed in crewmembers and controls 4 months before, immediately after, and 6 months after the mission. During the mission, crewmembers will wear a wrist-watch like device that measures movement activity and proximity to other devices 24/7 to investigate sleep-wake behavior and crew cohesion. Once monthly, subjects will perform the Cognition test battery to quantify changes in cognitive performance. Cognition was specifically designed for high-aptitude astronauts and astronaut surrogates. It consists of 10 brief, validated neuropsychological tests that cover a wide range of cognitive domains. A 24-hour, two-electrode electrocardiogram (ECG) will be performed monthly to investigate systematic changes in heart rate, heart rate variability, objectively assessed workload, and sleep fragmentation with time-in-mission. Behavioral alertness will be assessed with a 3 min. Psychomotor Vigilance Test (PVT) on a weekly basis along with brief surveys to assess subjective ratings of mood, workload, stress, sleep quality, tiredness, sickness, and conflicts among crewmembers. The results will be compared with findings from Mars520 and International Space Station (ISS), as many of the variables to be gathered overlap with those successfully obtained by our team in these and other space analog environments. The Cognition test battery was also implemented in the Antarctic stations Neumayer-III and Halley-VI. After this project we will have a much better understanding whether, to what extent, and for how long neurostructural and neurofunctional changes are induced in subjects over-wintering in the isolated and confined space analog environment of Concordia station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With the proposed work we will relevantly contribute to the goal of the Human Research Program (HRP) to provide human health and performance countermeasures, knowledge, technologies, and tools to enable safe, reliable, and productive human space exploration. More specifically, our findings, based on state-of-the-art neuroimaging technologies and on innovative, non-invasive, low burden, yet methodologically sound measurement technologies for cognitive, physiological, and crew cohesion outcomes, will relevantly contribute to the development of technologies to provide mission planners and system developers with strategies for monitoring and mitigating crew health and performance risks. These methodologies will also be useful for assessing subjects living in isolated, confined, and extreme environments on Earth.
Task Progress:	Data acquisition in the 2 Concordia winter-over crews has concluded. In January 2018, we received the final scans (6-month post-mission equivalent) of the second control group studied at DLR (German Aerospace Center) Cologne. Data acquisition for the whole study is therefore complete. Our efforts in recent months have thus mainly concentrated on the analysis of complex neuroimaging, actigraphy, proximity, ECG, and cognitive data. Preliminary neuroimaging results were presented at the 2018 NASA Human Research Program Investigators' Workshop in Houston, TX.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Boland EM, Rao H, Dinges DF, Smith RV, Goel N, Detre JA, Basner M, Sheline YI, Thase ME, Gehrman PR. "Meta-analysis of the antidepressant effects of acute sleep deprivation." J Clin Psychiatry. 2017 Sep/Oct;78(8):e1020-e1034. https://doi.org/10.4088/JCP.16r11332 ; PubMed PMID: 28937707 , Sep-2017
Articles in Peer-reviewed Journals	Moore TM, Basner M, Nasrini J, Hermosillo E, Kabadi S, Roalf DR, McGuire S, Ecker AJ, Ruparel K, Port AM, Jackson CT, Dinges DF, Gur RC. "Validation of the Cognition Test Battery for spaceflight in a sample of highly educated adults." Aerosp Med Hum Perform. 2017 Oct 1;88(10):937-46. https://doi.org/10.3357/AMHP.4801.2017 ; PubMed PMID: 28923143 , Oct-2017